

Towards Terabytes

Physics challenges the future of the hard drive industry, but manufacturers have foreseen it all. Here's an in-depth look at the technologies that will let you own a terabyte hard disk for your PC as early as next year

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Imagine a Boeing 747 flying at 600 mph four feet above the ground, trying to count blades of grass as it flies by. That's pretty much what your hard drive's head does—be it something as simple as opening a file in Notepad, or something as disk-intensive as file indexing! The blades of grass are, of course, magnetised areas on the disk platter. And these magnetised areas (bits) are being packed more closely together by disk manufacturers. That's why we keep seeing increased capacity hard disks.

In 1973, IBM shipped the model 3340 Winchester sealed hard disk drive, the predeces-

sor of all current hard disk drives. The 3340 had two spindles, each with a capacity of 30 MB. The Winchester got its name because it was invented at IBM's Winchester, NY, laboratory. What we use today are basically Winchester disks.

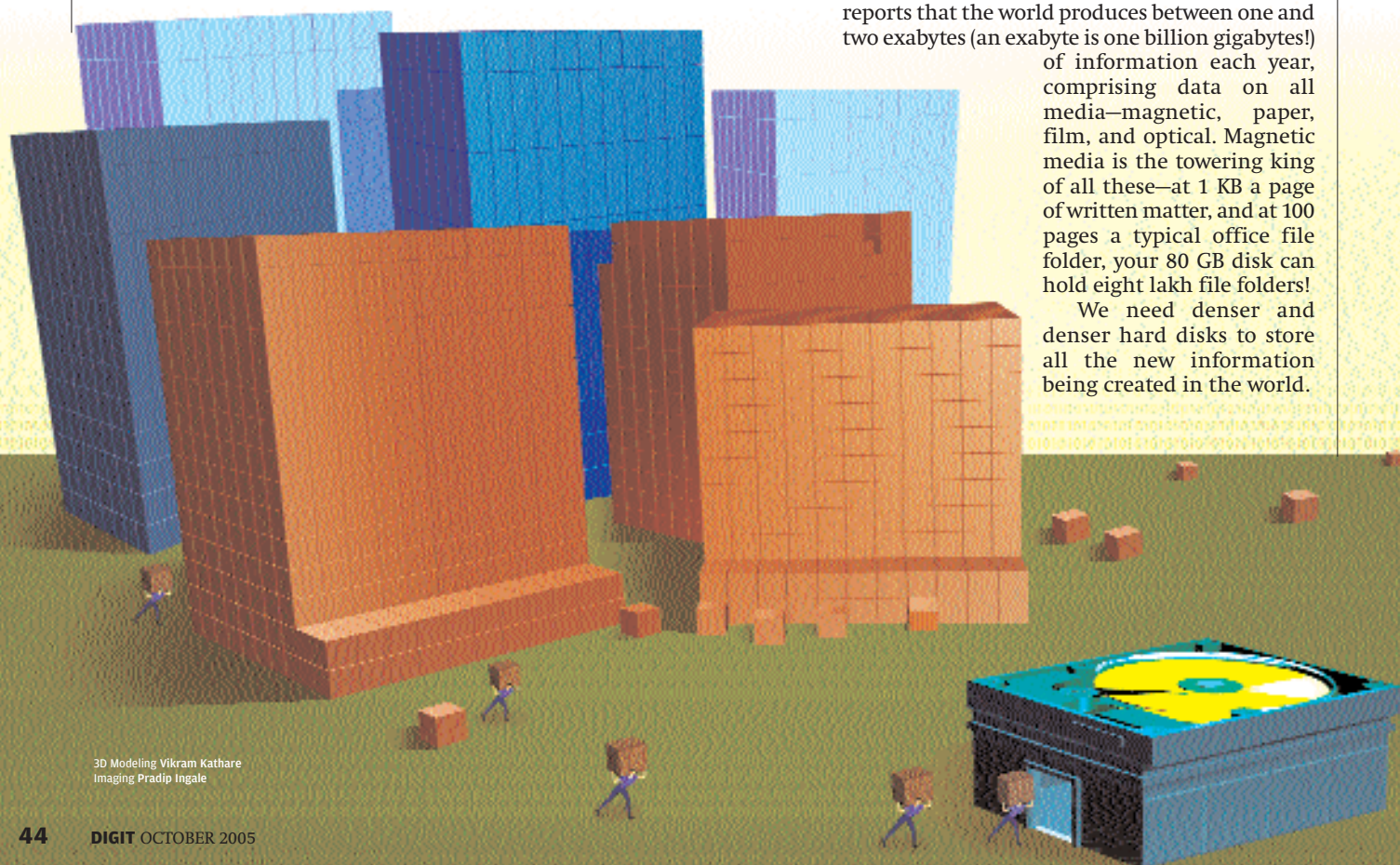
The recording method that's used with hard disks today dates back to the Winchester and beyond, and is called longitudinal recording (LR). And, as it turns out, LR is running out of steam. There is a method, though—namely, using AFC (AntiFerromagnetically-coupled) media—which promises to extend its life a little, but beyond that, we're on to other methods.

Who Needs Higher Data Densities?

The world does. A recent UC Berkeley study reports that the world produces between one and two exabytes (an exabyte is one billion gigabytes!)

of information each year, comprising data on all media—magnetic, paper, film, and optical. Magnetic media is the towering king of all these—at 1 KB a page of written matter, and at 100 pages a typical office file folder, your 80 GB disk can hold eight lakh file folders!

We need denser and denser hard disks to store all the new information being created in the world.



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